

Using Virtual Reality to Raise Awareness of Communication Challenges Faced by Individuals with Hearing Loss

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Figure 1: Hearing loss and hearing aid VR simulation demo using [16]

ABSTRACT

Hearing loss is typically managed with hearing aids, which amplify sound based on the user's hearing profile. However, unlike glasses for vision, hearing aids cannot address the reduced time-frequency resolution of the auditory system. To raise awareness of the challenges faced by hearing aid users, particularly in complex acoustic environments, we use virtual reality (VR). We integrate hearing loss and hearing aid simulators into a VR application, allowing users to experience the altered soundscape in an immersive way.

Index Terms: Hearing loss, social inclusion, immersive audio

1 INTRODUCTION

Hearing loss is one of the most prevalent health issues in modern society. The majority of hearing difficulties arise from damage to the cochlea, which cannot be cured or reversed by current medical procedures. In most cases, hearing impairment is managed with hearing aids—assistive devices that capture environmental sounds, amplify them, and deliver the processed signal directly to the user's ear. The amplification provided by a hearing aid is tailored to the user's hearing loss, usually diagnosed using audiometry—a hearing test that evaluates how well a person perceives sounds across various frequencies. The result of this test, known as an audiogram, maps the degree of hearing loss at different frequencies and serves as the basis for determining the level of amplification needed for each frequency to make otherwise imperceptible sounds audible. While it may seem intuitive that a hearing aid precisely matching the audiogram should fully restore hearing—just as glasses restore vision—this is a misconception. Hearing aids can make sounds louder and hence more accessible to the hearing-impaired, but they cannot recover the precision of a healthy auditory system (e.g., the

temporal and frequency resolution with which the auditory system processes sounds) [10]. This limitation becomes especially apparent in complex acoustic environments, such as restaurants, offices, or busy streets [12]. Consequently, hearing aids offer only limited benefits in such scenarios, often leaving users frustrated. In extreme cases, this frustration leads to users abandoning their devices or withdrawing from social interactions altogether.

Just like the small devices hidden behind the ear, the challenges faced by individuals with hearing impairments are often invisible to the normal-hearing population. Most people with normal hearing are unaware of their potential impact on communication barriers and may inadvertently make interactions more difficult for the hearing-impaired. Without understanding the nature of hearing loss, they seldom feel responsible or empowered to adapt their behavior. Hearing loss simulators offer an interesting solution for raising awareness about these challenges. Such simulators are algorithms that distort sound to mimic the effects of hearing loss. When a person with normal hearing uses a hearing loss simulator, their hearing test results resemble those of a hearing-impaired individual. Hearing loss simulators have a long history in research, with numerous versions developed and used in studies over the years [8, 16, 13, 15].

It is impossible to fully place oneself in another person's shoes, and a normal-hearing individual cannot truly experience life with hearing loss. However, recent advancements in virtual reality (VR) have demonstrated its effectiveness as a tool for educating people about the experiences of others [1]. For example, [9] found that individuals who experienced in VR what it would be like to lose their jobs and homes developed a deeper and more lasting compassion for the homeless. This approach to fostering empathy is also gaining interest in the context of hearing impairments [6, 4, 14]. Recently, a 3DTI real-time hearing loss simulator with VR integration support was developed by [16], offering the possibility to simulate hearing loss across various immersive acoustic scenarios. Despite the potential of this powerful tool and the growing interest in its use for research, there are few public demonstrations of the technology, aside from academic papers, that could effectively promote social

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inclusion.

With this demonstration, we aim to contribute to a broader understanding of the communication challenges faced by individuals with hearing loss. We present a VR experience that replicates one of the most common and difficult acoustic scenarios—a conversation at a table. By integrating a hearing loss simulator into the virtual environment, we enable users to experience varying degrees and types of hearing loss, as well as the effects of corresponding hearing aids. This immersive experience helps participants better understand the impact of hearing loss, the functionality of hearing aids, and the extent to which these devices can improve auditory perception in real-world situations.

2 VR DEMO

We developed a virtual reality (VR) application using Unity software to simulate a challenging acoustic scenario: a conversation between four individuals at a restaurant table. In the VR scene, the user is seated at the table alongside four virtual characters. The user experiences the scene by listening to the conversation at the table. After an initial introduction, the user selects one of three hearing loss profiles to experience: mild, moderate, or severe hearing loss. Once the hearing loss is applied, the user observes the effects on the auditory scene, noting that certain portions of the audio become inaudible. Subsequently, the user can choose to activate a hearing aid corresponding to the selected hearing loss profile. While the hearing aid improves audibility, allowing more of the scene to be heard and understood, the sound remains distorted, and separating speech from background noise is still challenging. This demonstrates how a hearing-impaired person can benefit from a hearing aid but at the same time highlights the limitations of hearing devices.

Each virtual character was assigned an audio source extracted from the DiPCo dataset, which provides close-microphone recordings of real conversations [18]. Notably, this dataset contains individual monaural recordings for each participant in the conversation. This feature enables individual spatialization of each character's voice in the VR environment, even when they speak simultaneously. The restaurant noise is simulated using an ambisonics recording obtained from Freesound [7]. Real-time sound source spatialization and room acoustic simulations are achieved using the open-source Steam Audio library [3] and the sound is delivered to the user binaurally using headphones. The hearing loss and hearing aid simulators were integrated into the Unity scene as custom audio mixer plugins with multiple exposed parameters. The parameters were dynamically modified via C scripts in response to changes in the hearing loss or hearing aid settings. The audiograms for the three hearing loss profiles were based on the most common hearing loss patterns described in [2]. The hearing aid correction strategy was implemented using the Fig6 fitting algorithm [11], as provided by the 3DTI software [5].

3 SUMMARY

In this brief report, we described our VR demo, which allows users to experience simulated hearing loss and hearing aid in a realistic, complex acoustic scenario. Our objective was to develop a demonstration that brings a greater understanding among normal-hearing individuals and encourage them to adopt more inclusive behaviors when interacting with people with hearing impairments. Future research could investigate whether such VR-based simulations can improve the ability of normal-hearing users to adapt their behavioral strategies to better accommodate the needs of people with hearing loss. Additionally, the hearing loss simulation has a practical advantage: it can be easily incorporated as a processing condition for normal-hearing participants in studies of auditory perception within virtual reality. This provides an efficient approach to obtaining preliminary insights into the perceptual challenges faced by individuals with hearing impairment [17]

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